

Week 9: Study Sheet

EE0849 Introduction to Robotics — Mobile Robot Kinematics

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1 Configuration and Planar Pose

1.1 Problem 1.1

A wheeled robot has configuration $q = (x, y, \theta) = (3 \text{ m}, 1 \text{ m}, 60^\circ)$.

- a) Write the unit vector pointing in the robot's heading direction.
- b) If the robot drives **2 m forward**, what is its new configuration?
- c) If the robot instead **rotates in place** by $+30^\circ$, what is its new configuration?

1.1.1 Solution

a) **Heading unit vector:**

$$\hat{f} = (\cos \theta, \sin \theta) = (\cos 60^\circ, \sin 60^\circ) = (0.5, 0.866)$$

b) **Drive 2 m forward.** The displacement is $d \cdot \hat{f}$, so:

$$x_{\text{new}} = 3 + 2(0.5) = 4.0 \text{ m}, \quad y_{\text{new}} = 1 + 2(0.866) = 2.732 \text{ m}, \quad \theta_{\text{new}} = 60^\circ$$
$$q_{\text{new}} = (4.0, 2.732, 60^\circ)$$

c) **Rotate in place by $+30^\circ$** (no translation):

$$q_{\text{new}} = (3.0, 1.0, 90^\circ)$$

1.2 Problem 1.2

A unicycle starts at $q(0) = (0, 0, 0)$ and is commanded with constant inputs $v = 0.5 \text{ m/s}$ and $\omega = 0.2 \text{ rad/s}$. Find the configuration $q(t)$ at $t = 2 \text{ s}$ by integrating the unicycle kinematics.

1.2.1 Solution

The unicycle equations are:

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \omega$$

Because ω is constant, integrate θ first:

$$\theta(t) = 0 + \omega t = 0.2t \quad \Rightarrow \quad \theta(2) = 0.4 \text{ rad} \approx 22.9^\circ$$

Then integrate x and y . With $\omega \neq 0$, the closed-form integrals are:

$$x(t) = \frac{v}{\omega}(\sin \theta(t) - \sin \theta(0)), \quad y(t) = -\frac{v}{\omega}(\cos \theta(t) - \cos \theta(0))$$

Plugging in at $t = 2 \text{ s}$:

$$x(2) = \frac{0.5}{0.2}(\sin 0.4 - 0) = 2.5 \times 0.3894 = 0.974 \text{ m}$$
$$y(2) = -\frac{0.5}{0.2}(\cos 0.4 - 1) = -2.5 \times (-0.0789) = 0.197 \text{ m}$$

$$q(2) = (0.974 \text{ m}, 0.197 \text{ m}, 0.4 \text{ rad})$$

Sanity check: the robot is tracing a circle of radius $R = v/\omega = 2.5 \text{ m}$. It has covered an arc angle of $\omega t = 0.4 \text{ rad}$ (22.9°) along that circle.

2 Holonomic vs Non-Holonomic Constraints

2.1 Problem 2.1

Classify each of the following constraints as **holonomic** or **non-holonomic**, and explain why:

- a) A train on a straight rail at $y = 5$: $y = 5$.
- b) A pendulum of length L swinging in 2D: $x^2 + y^2 = L^2$.
- c) A rolling coin on a plane: $\dot{x} \sin \theta - \dot{y} \cos \theta = 0$.

2.1.1 Solution

a) Holonomic. The constraint $y = 5$ involves only position variables (no velocities). It reduces the configuration space from 2D to 1D — the train can never be anywhere except on the rail.

b) Holonomic. The constraint $x^2 + y^2 = L^2$ is a pure position relationship with no velocity terms. The pendulum bob is restricted to a 1D circle in the plane.

c) Non-holonomic. The constraint involves velocities $(\dot{x}, \dot{y})$ and the heading θ , and it *cannot* be integrated to a pure position relationship $f(x, y, \theta) = 0$. The rolling coin can still reach any (x, y, θ) in the plane given enough maneuvering — only the *instantaneous* velocity direction is restricted.

2.2 Problem 2.2

A wheeled robot has heading $\theta = 45^\circ$ and instantaneous velocity $(\dot{x}, \dot{y}) = (1.0, 1.0)$ m/s. Does this velocity satisfy the rolling (no-slip) constraint $\dot{x} \sin \theta - \dot{y} \cos \theta = 0$?

2.2.1 Solution

Plug in:

$$\dot{x} \sin 45^\circ - \dot{y} \cos 45^\circ = 1.0(0.707) - 1.0(0.707) = 0 \quad \checkmark$$

Yes — the velocity is exactly along the heading direction $(\cos 45^\circ, \sin 45^\circ) = (0.707, 0.707)$, so it has zero sideways component. The constraint is satisfied.

3 The Unicycle Model

3.1 Problem 3.1

Write down the unicycle kinematics $\dot{q} = G(q)u$ with $u = (v, \omega)$, and show that the no-side-slip constraint $\dot{x} \sin \theta - \dot{y} \cos \theta = 0$ is automatically satisfied.

3.1.1 Solution

The unicycle model is:

$$\dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix}$$

So $\dot{x} = v \cos \theta$ and $\dot{y} = v \sin \theta$. Substituting into the constraint:

$$\dot{x} \sin \theta - \dot{y} \cos \theta = v \cos \theta \sin \theta - v \sin \theta \cos \theta = 0 \quad \checkmark$$

The unicycle model is built from the null space of the rolling constraint, so it satisfies the constraint by construction.

3.2 Problem 3.2

A unicycle is commanded with:

- a) $v = 1$ m/s, $\omega = 0$
- b) $v = 0$, $\omega = 1$ rad/s
- c) $v = 1$ m/s, $\omega = 0.5$ rad/s

For each case, describe the resulting trajectory and give the turning radius (if applicable).

3.2.1 Solution

- a) Pure forward motion. The robot drives in a straight line along its current heading. Turning radius: $R = \infty$.
- b) Pure rotation in place. The robot spins without translating ($\dot{x} = \dot{y} = 0$). Turning radius: $R = 0$.
- c) Circular motion. Turning radius:

$$R = \frac{v}{\omega} = \frac{1.0}{0.5} = 2 \text{ m}$$

The robot traces a circle of radius 2 m centered at a point 2 m to the left of its initial heading (for positive ω).

4 Differential-Drive Mapping

4.1 Problem 4.1

A differential-drive robot has wheel radius $r = 0.04$ m and track width $L = 0.20$ m. Its right and left wheel angular speeds are $\dot{\phi}_R = 12$ rad/s and $\dot{\phi}_L = 8$ rad/s.

- a) Compute the linear speed v and the angular speed ω of the robot.
- b) Compute the turning radius.
- c) Sketch the robot's trajectory.

4.1.1 Solution

a) **Forward kinematics:**

$$v = \frac{r}{2}(\dot{\phi}_R + \dot{\phi}_L) = \frac{0.04}{2}(12 + 8) = 0.02 \times 20 = 0.40 \text{ m/s}$$

$$\omega = \frac{r}{L}(\dot{\phi}_R - \dot{\phi}_L) = \frac{0.04}{0.20}(12 - 8) = 0.2 \times 4 = 0.80 \text{ rad/s}$$

b) **Turning radius:**

$$R = \frac{v}{\omega} = \frac{0.40}{0.80} = 0.50 \text{ m}$$

c) **Trajectory:** a circle of radius 0.5 m. Because $\dot{\phi}_R > \dot{\phi}_L$, the robot curves to the **left** (positive ω).

4.2 Problem 4.2

The same robot ($r = 0.04$ m, $L = 0.20$ m) is commanded to drive forward at $v = 0.30$ m/s while turning at $\omega = 1.5$ rad/s. Find the required wheel speeds $\dot{\phi}_R$ and $\dot{\phi}_L$.

4.2.1 Solution

Invert the differential-drive mapping:

$$\dot{\phi}_R = \frac{v + \omega L/2}{r}, \quad \dot{\phi}_L = \frac{v - \omega L/2}{r}$$

Plug in:

$$\begin{aligned} \dot{\phi}_R &= \frac{0.30 + 1.5 \times 0.10}{0.04} = \frac{0.30 + 0.15}{0.04} = \frac{0.45}{0.04} = 11.25 \text{ rad/s} \\ \dot{\phi}_L &= \frac{0.30 - 1.5 \times 0.10}{0.04} = \frac{0.30 - 0.15}{0.04} = \frac{0.15}{0.04} = 3.75 \text{ rad/s} \end{aligned}$$

Sanity check: average wheel speed = $(11.25 + 3.75)/2 = 7.5$ rad/s, and $v = r \cdot 7.5 = 0.30$ m/s.

4.3 Problem 4.3

For the differential-drive robot from Problem 4.1, what is the **maximum** forward speed if each motor can spin at most $\dot{\phi}_{\max} = 20$ rad/s? What is the **maximum turn rate** (while stationary)?

4.3.1 Solution

Maximum forward speed: both wheels at maximum, same direction.

$$v_{\max} = r \cdot \dot{\phi}_{\max} = 0.04 \times 20 = 0.80 \text{ m/s}$$

Maximum turn rate in place: wheels at maximum in opposite directions ($\dot{\phi}_R = +20$, $\dot{\phi}_L = -20$, so $v = 0$).

$$\omega_{\max} = \frac{r}{L}(\dot{\phi}_R - \dot{\phi}_L) = \frac{0.04}{0.20}(20 - (-20)) = 0.2 \times 40 = 8.0 \text{ rad/s}$$

5 The Bicycle Model

5.1 Problem 5.1

A car is modeled by the bicycle model with wheelbase $L = 2.5$ m and maximum steering angle $\delta_{\max} = 30^\circ$.

- a) What is the **minimum turning radius**?
- b) At $v = 5$ m/s and $\delta = 10^\circ$, compute $\dot{\theta}$ and the current turning radius.

5.1.1 Solution

a) **Minimum turning radius:**

$$R_{\min} = \frac{L}{\tan \delta_{\max}} = \frac{2.5}{\tan 30^\circ} = \frac{2.5}{0.577} \approx 4.33 \text{ m}$$

b) Angular velocity and radius at $\delta = 10^\circ$:

$$\dot{\theta} = \frac{v \tan \delta}{L} = \frac{5.0 \times \tan 10^\circ}{2.5} = \frac{5.0 \times 0.1763}{2.5} = 0.353 \text{ rad/s}$$

$$R = \frac{L}{\tan \delta} = \frac{2.5}{0.1763} = 14.18 \text{ m}$$

Consistency check: $R = v/\dot{\theta} = 5.0/0.353 = 14.18$ m.

5.2 Problem 5.2

Why can't a car-like (bicycle-model) robot turn in place, while a differential-drive robot can?

5.2.1 Solution

The bicycle model's heading rate is $\dot{\theta} = v \tan \delta / L$. This is **coupled to** v : if $v = 0$, then $\dot{\theta} = 0$ regardless of the steering angle δ . Physically, the rear wheel must roll (move the body forward or backward) for the geometry to rotate the body.

The differential-drive model has $\dot{\theta} = \omega$ as an *independent* control input. Setting $\dot{\phi}_R = +\Omega$ and $\dot{\phi}_L = -\Omega$ (opposite directions) yields $v = 0$ and $\omega = r\Omega \cdot 2/L \neq 0$. The robot rotates in place without translating — because the two wheels push in opposite directions, cancelling translation while accumulating rotation.

This is the fundamental advantage of differential drive in tight indoor environments.

6 Instantaneous Center of Rotation (ICR)

6.1 Problem 6.1

A differential-drive robot has track width $L = 0.50$ m. Its wheels are moving at $v_R = 0.6$ m/s and $v_L = 0.2$ m/s.

- a) Compute ω and the turning radius R measured from the center of the axle.
- b) How far from the **left wheel** is the ICR?
- c) What is the linear speed of a point at the center of the robot?

6.1.1 Solution

a)

$$\omega = \frac{v_R - v_L}{L} = \frac{0.6 - 0.2}{0.5} = 0.80 \text{ rad/s}$$

$$v_{\text{centre}} = \frac{v_R + v_L}{2} = 0.40 \text{ m/s}$$

$$R = \frac{v_{\text{centre}}}{\omega} = \frac{0.40}{0.80} = 0.50 \text{ m}$$

b) Distance from left wheel to ICR. The ICR lies on the line through the axle, and the left (slower) wheel is *closer* to it. The left wheel is $L/2 = 0.25$ m from the robot center, on the same side as the ICR. So:

$$d_{L \rightarrow \text{ICR}} = R - L/2 = 0.50 - 0.25 = 0.25 \text{ m}$$

Cross-check with kinematics: at the left wheel, $v_L = \omega \cdot d_{L \rightarrow \text{ICR}} = 0.80 \times 0.25 = 0.20$ m/s.

c) Center speed: already computed, $v_{\text{centre}} = 0.40$ m/s, which equals $\omega \cdot R = 0.80 \times 0.50$.

6.2 Problem 6.2

A car (bicycle model) has wheelbase $L = 3.0$ m and is driving at $v = 10$ m/s on a circular path of radius $R = 20$ m.

- a) Find the steering angle δ .
- b) Find the heading rate $\dot{\theta}$.
- c) How long does it take to complete one full lap?

6.2.1 Solution

a) **Steering angle:**

$$\delta = \arctan\left(\frac{L}{R}\right) = \arctan\left(\frac{3.0}{20}\right) = \arctan(0.15) \approx 8.53^\circ$$

b) **Heading rate:**

$$\dot{\theta} = \frac{v}{R} = \frac{10}{20} = 0.50 \text{ rad/s}$$

c) **Lap time:**

$$T = \frac{2\pi}{\dot{\theta}} = \frac{2\pi}{0.50} \approx 12.57 \text{ s}$$

Equivalently: circumference $= 2\pi R = 125.66$ m, at $v = 10$ m/s gives $T = 12.57$ s.

7 Mecanum / Omnidirectional Motion

7.1 Problem 7.1

Why is a mecanum-wheeled robot called **holonomic**, while a differential-drive robot is **non-holonomic**, even though both live in the same 3-D configuration space (x, y, θ) ?

7.1.1 Solution

The distinction is about how many *independent* instantaneous velocities the robot can command.

- A **differential-drive** robot has 2 control inputs (v, ω) mapped to a 3-D velocity $\dot{q} = (\dot{x}, \dot{y}, \dot{\theta})$. The sideways velocity component $\dot{x} \sin \theta - \dot{y} \cos \theta$ is always zero — one velocity direction is forbidden. This 2-out-of-3 gap is the non-holonomic constraint.
- A **mecanum-wheel** robot has 4 wheel control inputs that, after the diagonal roller geometry is accounted for, map onto a full 3-D velocity $(\dot{x}, \dot{y}, \dot{\theta})$ with no forbidden directions. 3 controls for 3 DOF $\rightarrow$ no velocity constraint $\rightarrow$ **holonomic**.

In practice this means a mecanum robot can translate sideways or diagonally *without* first rotating — something a differential-drive robot physically cannot do.

7.2 Problem 7.2

A four-mecanum robot is commanded with the wheel speeds below (sign convention: positive means the wheel rolls forward relative to the body heading):

Wheel	Speed
Front-left (FL)	+1
Front-right (FR)	-1
Back-left (BL)	-1
Back-right (BR)	+1

What motion does the robot produce? Give a qualitative answer based on the mecanum motion patterns from class.

7.2.1 Solution

Diagonally opposite wheels have the same sign (FL and BR both +1; FR and BL both -1). Adjacent wheels have opposite signs. Referring to the motion table from class, this is the **strafe right** command — the robot translates sideways to the right without rotating and without changing its heading.

Intuition: the FL and BR rollers (at $+45^\circ$) both push the body to the right as they roll forward, while the FR and BL rollers (at -45°) also push to the right as they roll backward. The body accelerations in the x -direction add; the y - and θ -components cancel.

8 Summary of Key Formulas

Unicycle model

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \omega$$

Differential-drive (forward)

$$v = \frac{r}{2}(\dot{\phi}_R + \dot{\phi}_L), \quad \omega = \frac{r}{L}(\dot{\phi}_R - \dot{\phi}_L)$$

Differential-drive (inverse)

$$\dot{\phi}_R = \frac{v + \omega L/2}{r}, \quad \dot{\phi}_L = \frac{v - \omega L/2}{r}$$

Turning radius

$$R = \frac{v}{\omega}$$

Bicycle model

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \frac{v \tan \delta}{L}$$

Turning radius (bicycle)

$$R = \frac{L}{\tan \delta}, \quad R_{\min} = \frac{L}{\tan \delta_{\max}}$$

Rolling / no-slip constraint

$$\underbrace{v = r\dot{\phi}}_{\text{rolling}}, \quad \underbrace{\dot{x} \sin \theta - \dot{y} \cos \theta = 0}_{\text{no lateral slip}}$$